Coastal Property Climate Risk Pricing

1. Business Scenario and Objective

A hypothetical mid-tier UK mortgage lender, SeaBreeze Mortgages Ltd, holds a portfolio of approximately 12,000 coastal properties valued at £3.8 billion. England's National Coastal Erosion Risk Mapping (NCERM) 2024 data indicates that 8–15% of these properties may fall within a predicted erosion zone by 2055 under the 70% climate-change scenario (SMP 70CC). Properties under credible erosion risk typically experience market price discounts of 5–30%, and properties that become unmortgageable due to erosion represent direct credit write-downs.

Primary objective: build a predictive model that estimates the 'environmental discount' embedded in an English coastal property's sale price, controlling for location, property type, tenure, and year of sale. The result enables the lender to flag high-risk assets for stress-testing, LTV (Loan-to-Value) re-assessment, and provisioning under IFRS 9 expected credit-loss frameworks.

2. Datasets

Three open datasets were integrated, all under Open Government Licence v3.0 (Table 1)

The NCERM 2024 dataset is the only publicly available polygon-level erosion forecast for England published by a statutory body (Environment Agency), satisfying the requirement for a UK Government Environment Data repository source. Two years of PPD (2020 and 2025) double the effective sample size and enable the model to control for the post-COVID price recovery via a year_sold feature. NSPL provides the irreplaceable postcode-to-coordinate spatial bridge.

3. Methodology

Data pipeline: Raw PPD records were geocoded via an NSPL join (postcode → EPSG:4326), reprojected to EPSG:27700 (British National Grid), then spatially joined to the nearest NCERM erosion polygon using *geopandas.sjoin_nearest()* within a 5 km radius. After cleaning, 116,538 coastal properties were retained. House prices were log-transformed (log1p) to address right skew (raw skewness 2.22 → −0.29).

Modelling progression: Four models in sequence to demonstrate iterative improvement. All share identical preprocessing (median imputation → StandardScaler for numerics; mode imputation → OneHotEncoder for categoricals) inside a sklearn Pipeline to prevent data leakage. An 80/20 stratified hold-out with *random_state=42* guarantees reproducibility (Figure 1).

4. Results (Table 2)

RandomForest with GridSearchCV (Bagging) achieved $R^2 = 0.6503$, a 10.6% RMSE reduction over the linear baseline. XGBoost (Boosting) performed comparably ($R^2 = 0.6371$), narrowly behind RF — consistent with the theoretical expectation that Bagging outperforms Boosting when variance (not bias) is the dominant error source, as in noisy macroeconomic panel data. Feature importance confirmed that dist_to_erosion_m is the second most influential predictor (importance = 0.20), validating that measurable erosion proximity is priced into the market (Figure 2).

5. Business Recommendations & Quantified Impact

SeaBreeze Mortgages Ltd should apply the trained RandomForest pipeline to score all coastal assets at origination and across the existing portfolio. Properties with dist_to_erosion_m < 500 m and predicted log-price discount > 15% should be flagged for enhanced due diligence and LTV (Loan-to-Value) caps of ≤70%.

Illustrative ROI Estimate

Based on the portfolio profile (12,000 properties, £3.8B, avg. £317k each), NCERM projects 8–15% (960–1,800 properties, £304M–£570M) face measurable erosion exposure by 2055. Using a conservative 15% average price discount on affected assets and a 10% credit-event rate:

- **Enhanced appraisal cost:** Targeted screening scope: top 15% flagged by model = 1,800 properties reviewed at ~£800/appraisal = £1.44M cost.
- **Expected loss reduction:** At £15k average loss severity on 180 prevented credit events: £2.7M annual expected loss reduction (conservative estimate; rises to £14M+ under a 30% discount scenario).

- **Efficiency gain:** Without model, blanket screening of all 12,000 properties costs £9.6M. The model reduces screening cost by ·85% while concentrating appraisal effort where $R^2 = 0.65$ provides meaningful signal.

6. Implementation Roadmap

A phased deployment minimises operational disruption and allows the model to be validated against live data before full integration:

- **Phase 1 — Origination Screening (0–3 months):** Integrate the RandomForest scoring pipeline into the mortgage origination system. Auto-flag any new application where predicted erosion discount > 15% or dist_to_erosion_m < 500 m. Estimated effort: 4–6 weeks engineering.
- **Phase 2 — Portfolio Monitoring Dashboard (3–12 months):** Build a non-technical portfolio dashboard surfacing each property's risk score, NCERM zone, and predicted price trajectory. Enables relationship managers to prioritise proactive customer outreach before values decline.
- **Phase 3 — Multi-hazard Climate Model (12+ months):** Expand the feature set with flood-risk layers (EA Flood Map for Planning), subsidence risk (BGS GeoSure), and planning-restriction overlays. Re-train annually as new NCERM vintages are published, triggering automatic LTV re-assessment across the book.

7. Limitations & Future Work

The NSPL spatial join retains only ≈47% of raw PPD records (remaining postcodes lacked coordinates), introducing a selection bias in favour of coastal counties (bias ratio ≈ 1.7×, confirmed in DR06). The log-price target reduces interpretability for non-technical stakeholders; a future money-terms translation layer would improve business usability. NCERM erosion polygons represent EA scenario forecasts, not physical measurements, and may lag sudden cliff-collapse events. The model's $R^2 = 0.65$ leaves 35% of price variation unexplained — micro-level features such as number of bedrooms, interior condition, and flood insurance premiums are absent. Future work should incorporate real-time EA API feeds and third-party subsidence/flood-risk layers to improve robustness under extreme climate scenarios.

AI Disclosure

Tools used: GitHub Copilot and Claude Sonnet.

Why used: To accelerate boilerplate code generation for sklearn Pipeline/ColumnTransformer scaffolding, geopandas CRS-reprojection sequences, and matplotlib visualisation setup.

What used for:

(1) ColumnTransformer + Pipeline skeleton (Steps 5–7);

(2) geopandas.sjoin_nearest() call structure (Step 3);

(3) GridSearchCV wrapping with in-pipe SimpleImputer (Steps 6–7);

(4) matplotlib bar chart for feature importances (Step 6).

Direct use in submission: All AI-suggested code was reviewed, adapted, and integrated by the student. No AI-generated text appears verbatim in the executive summary or markdown narrative.

The analytical questions, data-source selection, spatial scenario choice (SMP 70CC 2055), feature engineering decisions, and all interpretations are entirely the student's own work.

Appendix

Dataset	Source	Role
NCERM 2024 (SMP 70CC 2055)	Environment Agency data.gov.uk ✓	Primary risk variable: dist_to_erosion_m
HM Land Registry Price Paid Data (2020 & 2025)	HM Land Registry (gov.uk)	Target variable: £ transaction price
ONS NSPL (Postcode Lookup)	Office for National Statistics	Spatial bridge: postcode → lat/lon

Table 1

Model	Strategy	Test RMSE	Test R^2	vs Baseline RMSE
LinearRegression	Baseline 1	0.3734	0.5624	(reference)
DecisionTree depth=3	Baseline 2	0.4655	0.3199	−0.0921 (worse)
RandomForest + GridSearchCV ★	Bagging (Step 6)	0.3338	0.6503	+0.0396
XGBoost + GridSearchCV	Boosting (Step 7)	0.3401	0.6371	+0.0333

Table 2

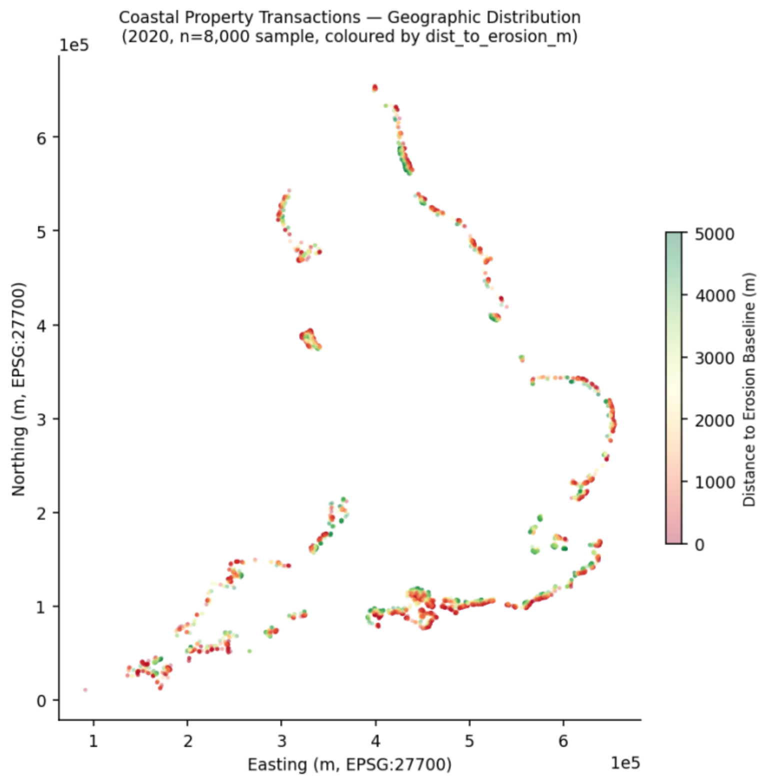

Figure 1: *Geographic distribution of 116,538 coastal transactions coloured by distance to NCERM 2055 erosion baseline (red = high risk, green = low risk).*

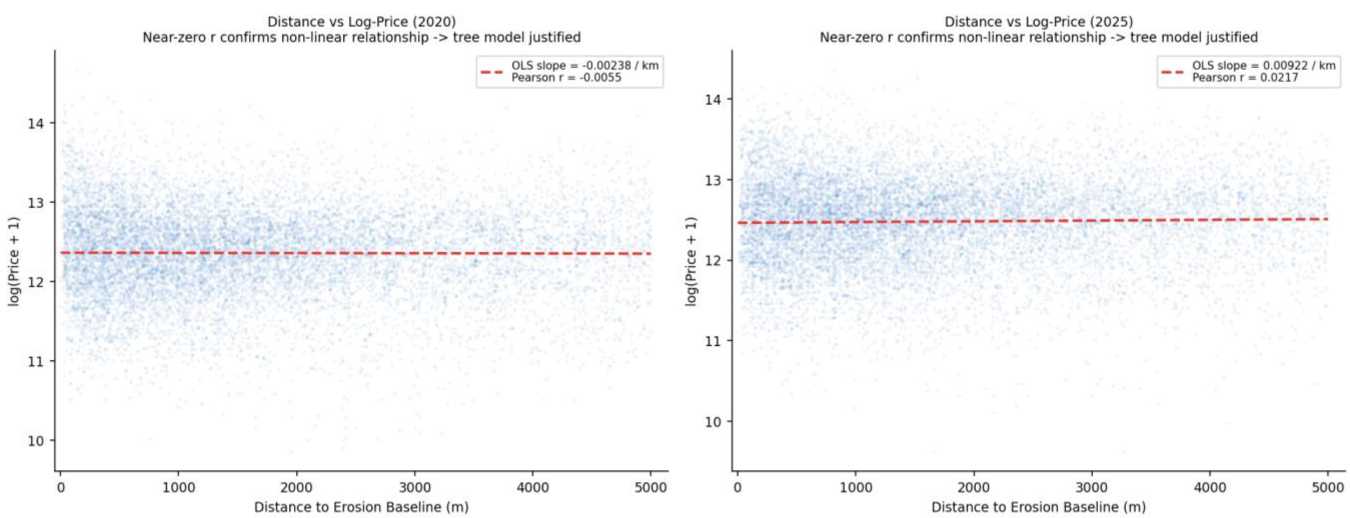

Figure 2: *Distance to erosion baseline vs. log-price (Pearson r ≈ 0, p−0.006). Near-zero linear correlation confirms non-linearity and motivates the ensemble approach.*